VIT Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Information Technology

Semester	T.E. Semester VI
Subject	Devops Lab
Laboratory	CC-02 Lab

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Experiment Number	01
Experiment Title	To understand DevOps Principles, Practices and DevOps Engineer Roles and Responsibilities.
Aim	To understand the core principles and practices of DevOps and the roles and responsibilities of a DevOps Engineer in the software development lifecycle.

Theory

What is DevOps?

DevOps is all about automating and streamlining the software development lifecycle so that code moves from development to production quickly, reliably, and securely.

DevOps Stages with Industry Tools

Build Stage: Source code is written and managed using version control systems like Git and GitHub/GitLab. CI tools such as Jenkins or GitLab CI/CD automatically compile, build, and package the application, while build tools like Maven and Gradle manage dependencies and generate deployable artifacts.

Test Stage: Automated testing ensures code quality and reliability. JUnit and TestNG handle unit and integration tests, Selenium supports UI testing, and SonarQube performs static code analysis to detect bugs, code smells, and security vulnerabilities early in the pipeline.

Release Stage: Applications are first deployed to staging and then production using containerization and orchestration tools like Docker and Kubernetes. Configuration management and deployment automation are handled by Ansible, Helm, or ArgoCD, enabling strategies such as blue-green, canary, and rolling deployments.

Continuous Feedback Loop: Production systems are continuously monitored using Prometheus and Grafana for metrics, ELK Stack for logs, and Datadog or New Relic for performance monitoring. Alerts and user feedback help teams quickly resolve incidents and continuously optimize future releases.

Principles of DevOps

Collaboration
Development and Operations teams work together throughout the software development lifecycle to avoid delays and miscommunication.

Automation
Repetitive tasks such as build, test, and deployment are automated to improve efficiency and reduce human errors.

Continuous Integration (CI)
Code changes are frequently integrated into a shared repository and automatically built and tested.

Continuous Delivery/Deployment (CD)
The application is automatically prepared and deployed to testing or production environments after successful builds.

Infrastructure as Code (IaC)
Infrastructure and configuration are managed using code, ensuring consistency across environments.

Continuous Testing
Automated testing is performed at every stage to ensure software quality and reliability.

Monitoring and Feedback
System performance and logs are continuously monitored to detect issues early and improve future releases.

Roles and Responsibilities of DevOps Engineer

Roles

1. Acts as a bridge between Development, QA, Operations, and Security teams.
2. Designs and manages CI/CD pipelines for automated build, test, and deployment.

	3. Ensures system reliability, scalability, and performance in production environments. **Responsibilities** 1. Source Code & CI Management: Manage version control systems (Git) and configure CI tools like Jenkins or GitLab CI/CD. 2. Build & Deployment Automation: Automate builds and deployments using tools such as Maven, Docker, Kubernetes, and Ansible. 3. Infrastructure Management: Implement Infrastructure as Code using tools like Terraform or CloudFormation to maintain consistent environments. 4. Monitoring & Logging: Monitor applications and infrastructure using Prometheus, Grafana, ELK Stack, or Datadog. 5. Security & Compliance: Integrate security checks in pipelines (DevSecOps) using tools like SonarQube and vulnerability scanners. 6. Incident Management & Troubleshooting: Respond to system failures, analyze root causes, and restore services quickly. 7. Continuous Improvement: Optimize pipelines, improve automation, and document processes based on feedback and metrics.

Conclusion	This lab experiment helped in understanding the core principles and practices of DevOps and the roles and responsibilities of a DevOps Engineer across the software development lifecycle.